Continuous Probability Distributions

MGMT 670: Business Analytics

Purdue University — Mitchell E. Daniels, Jr. School of Business

Continuous Random Variable

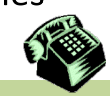
- Assumes any value in an interval or a collection of intervals on the real line
  - arrivals

What is the probability that the dean will call me exactly at 11:01 pm tonight?

- Is it possible to define the probability of the co... value?

What is the probability that the dean will call me between 11:00 and 11:05 pm?

- The probability for continuous random variables is defined for __an interval__.

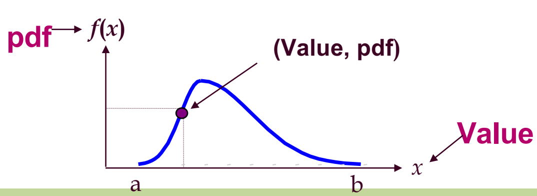


Continuous Probability Density Function

- Informally, a smoothed histogram, showing relative frequencies, *f*(*x*), of all values of the random variable, *X*.

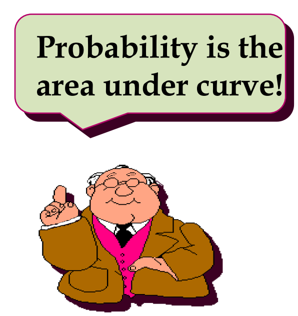
  However, *f(x) is not probability*.
- Properties of the probability density function (pdf)

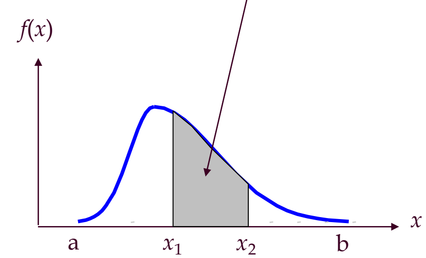
$$\int_{-\infty}^{+\infty} f(x)dx = 1, \quad f(x) \ge 0$$



Probability of Continuous Random Variable

$$P(x_1 \le X \le x_2) = \int_{x_1}^{x_2} f(x)dx$$



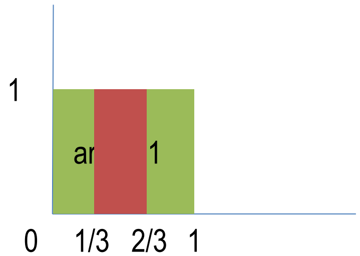


Properties of the probability density function

$$\int_{-\infty}^{+\infty} f(x)dx = 1, \quad f(x) \ge 0$$

Example: Uniform Distribution

Select a random number between 0 and 1



X is uniform random variable from the interval [0,1]

$f(x) = 1$ for all x in [0,1]

$$P(\frac{1}{3} \le X \le \frac{2}{3}) = ?$$

Normal Probability Distribution

- Describes many continuous random processes or phenomena.
- Can be used to approximate some discrete probability distributions such as Binomial.
- Basis for classical statistical inference.

Normal Probability Density Function

- Normal Probability Density Function (PDF)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Don't tattoo this!!!

where:

x = value of random variable ($-\infty \le x \le \infty$)

μ = mean

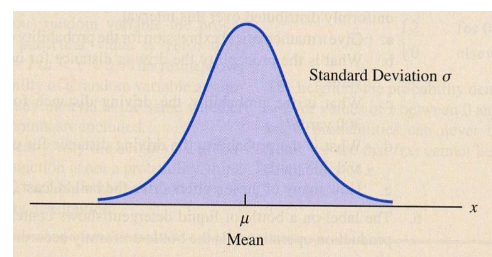
σ = standard deviation

π = 3.14159

e = 2.71828

Characteristics of Normal Probability Distribution

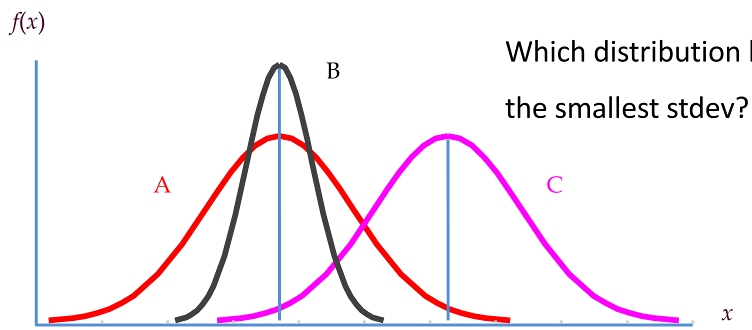
- The Normal Probability Density Function (PDF)



  - Bell-shaped and symmetric
  - Mean, median, and mode are equal.
  - Infinite Range

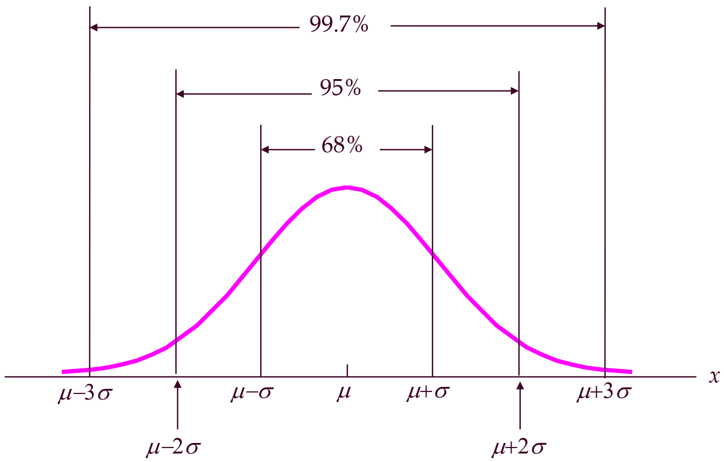
Parameters of Normal Probability Distribution

For a normal population, if we know the mean and standard deviation, we have the complete information about the population.



Which distribution has the smallest stdev?

Some Common Intervals of Normal Probability Distribution

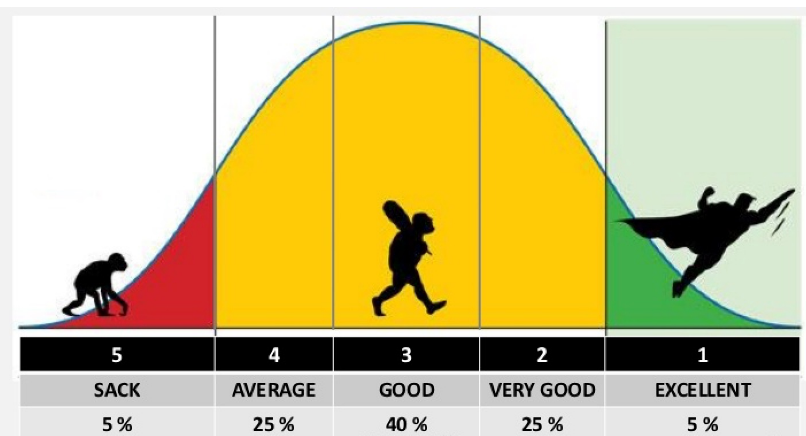


Why normal distributions?

Many distributions in real life

- Travel time by air plane from Chicago to NYC
- People heights
- Number of customers in a department store
- Number of daily visitors of a website
- …

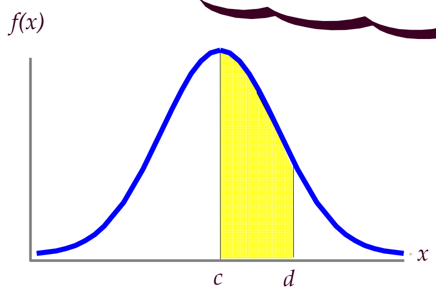
Employee Performance Evaluation

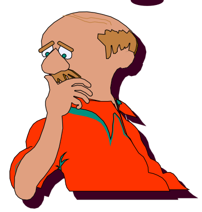


Normal Probability

- **Probability is the area under curve.**

$$P(c \le X \le d) = \int_c^d f(x)dx$$





Standard Normal Probability Distribution

- Normal distribution with a mean of zero and a standard deviation of one.
- *Z* is commonly used to designate the standardized normal random variable.
- Conversion rule

$$z = \frac{x - \mu}{\sigma}$$

- Measures the number of standard deviations the value *x* is from μ.

Example

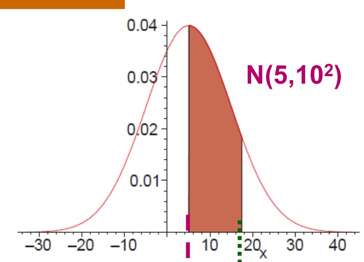
A junk stock's return is a normal distribution with mean 5, standard deviation 10 (in percentage).

We write its as either N(5,10²) or N(5,10)

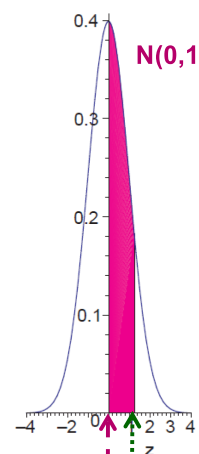
What is the probability that the return is between 5 and 17.5 this month?

Standardization Example

$P(5 \le X \le 17.5)$



$P(0 \le Z \le 1.25)$



$$z = \frac{5-5}{10} = 0.0$$

$$z = \frac{17.5-5}{10} = 1.25$$

Using Table to Find Normal Probabilities

- Standardize, then use standard normal table.

$$P(5 \le X \le 17.5) =$$

$$P\left(\frac{5-\mu}{\sigma} \le \frac{X-\mu}{\sigma} \le \frac{17.5-\mu}{\sigma}\right)$$

$$= P\left(\frac{5-5}{10} \le \frac{X-5}{10} \le \frac{17.5-5}{10}\right)$$

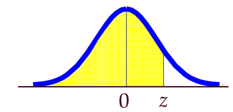
$$= P(0 \le Z \le 1.25)$$

$$= P(Z \le 1.25) - P(Z \le 0)$$ ➡ **Normal Z Table**

Cumulative Standard Normal Probability Table

$P(5 \le X \le 17.5) = P(0 \le Z \le 1.25)$

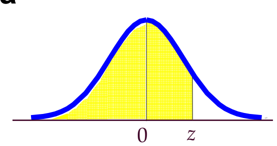
$= P(Z \le 1.25) - P(Z \le 0)$ $= .8944 - 0.5 = 0.3944$



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319

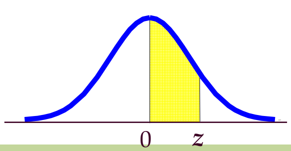
Finding Probabilities Using Normal Tables

- **Cumulative Standard Normal Table gives the shaded area**



$P(a \le Z \le b) = P(Z \le b) - P(Z \le a)$

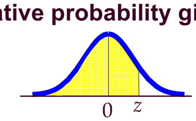
- **Standard normal table gives the following area under *z***



Using Excel Functions

- **Excel has two functions for computing probabilities for normal distributions:**
  - **NORM.S.DIST is used to compute the cumulative probability given a z value.**

    $P(Z \le z)$ = NORM.S.DIST(z, TRUE)



  - **NORM.DIST is used to compute the cumulative probability for x given a mean of μ and standard deviation of σ.**

    $P(X \le x)$ = NORM.DIST(x, μ, σ, TRUE)

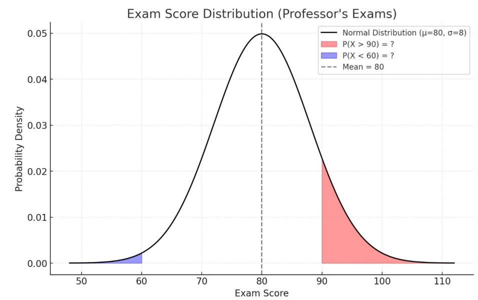
Using Excel Functions

- Using NORM.S.DIST
  $P(0 \le Z \le 1.25)$
  = NORM.S.DIST(1.25, TRUE) – NORM.S.DIST(0, TRUE)
  = 0.8944 – 0.5
  = 0.3944

- Using NORM.DIST
  $P(5 \le X \le 17.5)$
  = NORM.DIST(17.5,5,10,TRUE) – NORM.DIST(5,5,10,TRUE)
  = 0.8944 – 0.5
  = 0.3944

Exercise

- Your professor's exam scores are normally distributed with a mean of 80 and a standard deviation of 8. What's the probability that you'll score above 90? (Are you aiming too high?) And what's the probability that you'll score below 60?



Exercise

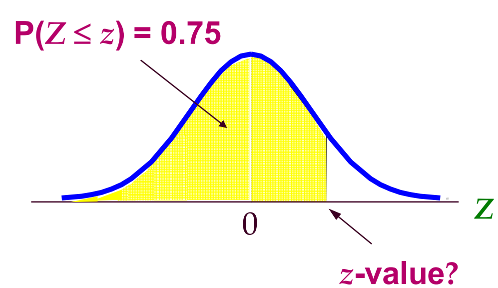
The flight time from Chicago O'Hare (ORD) to Purdue University (LAF) is approximately normally distributed with a mean of 60 minutes and a standard deviation of 8 minutes. What's the probability that a randomly selected flight has a duration:

Between 55 and 65 minutes?

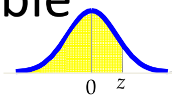
Longer than 70 minutes?

Why are these results useful?

Finding z-Value for a Known Cumulative Probability



Using Standard Normal Table



$P(Z \le z) = 0.75$ → $z \approx 0.675$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
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1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
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1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319

Using Excel Functions

- Excel has two functions for finding z- or x-value for normal distributions:
  - NORM.S.INV is used to find the z-value given a cumulative probability p.

    NORM.S.INV(p) = z such that $P(Z \le z) = p$

    z = NORM.S.INV(0.75) = 0.674

  - NORM.INV is used to find the x-value given a cumulative probability p, a mean of μ and standard deviation of σ.

    NORM.INV(p, μ, σ) = x such that $P(X \le x) = p$

Exercise

- Your total sleep time per night is normally distributed with a mean of 8 hours and a standard deviation of 1 hour. After several late nights during finals, you decide that you need to sleep at least as much as the top 10% of your normal nights. How many hours of sleep should you aim for? (Time to put your phone away!)

Exercise

The flight time from Chicago O'Hare (ORD) to Purdue University (LAF) is approximately normally distributed with a mean of 60 minutes and a standard deviation of 8 minutes. The middle 90% of flights will have durations between what two values?

Exercise

Pep Zone sells auto parts and supplies, including a popular multi-grade motor oil. The store uses a replenishment policy: place an order when the stock of this oil drops to 13 gallons. When an order is placed, it takes 1 day (the lead time) before the new shipment arrives. During this 1-day lead time, customer demand for the oil is normally distributed with mean 10 gallons and standard deviation 3 gallons.

a. What is the probability of a stock-out?

b. What should the reorder point be if Pep Zone wants the probability of a stock-out during the lead time to be only 5%?

Normal Approximation of Binomial Probabilities

- When the number of trials, *n*, becomes large the normal probability distribution provides an easy-to-use approximation of binomial probabilities

  where $n > 20$, $np \geq 5$, and $n(1 - p) \geq 5$.

- Set $\mu = np; \sigma = \sqrt{np(1-p)}$
- Add and subtract a **continuity correction factor** because a continuous distribution is being used to approximate a discrete distribution. For example,

  $P(x = 10)$ is approximated by $P(9.5 \leq x \leq 10.5)$.

Normal Approximation to the Binomial Distribution

When $X \sim \text{Binomial}(n, p)$, the exact distribution can be approximated by a Normal distribution when n is large and p is not too close to 0 or 1:

$$X \approx N\big(\mu = np,\ \sigma^2 = np(1-p)\big).$$

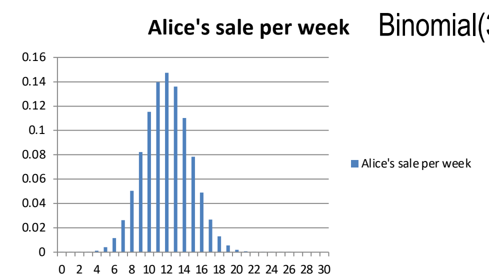
Because X is discrete and the Normal is continuous, we apply a **continuity correction**:

- To approximate $P(X = k)$, use $P(k - 0.5 < Y < k + 0.5)$, where $Y \sim N(\mu, \sigma^2)$.
- To approximate $P(X \leq k)$, use $P(Y < k + 0.5)$.
- To approximate $P(X \geq k)$, use $P(Y > k - 0.5)$.

Normal Approximation of Binomial Probabilities

For a binomial distribution Binomial(n,p), when the number of trials, n, becomes large:

Its distribution looks like normal distribution



Binomial(30,.4)

https://istats.shinyapps.io/BinomialDist/

Example

n = 100, *p* = .4. Find $P(30 \le x \le 50)$

- Using the binomial distribution, the result is?

Check: $np \ge 5$, $n(1-p) \ge 5$, $n \ge 20$.

- Using the normal approximation,

$P(29.5 \le x \le 50.5) = ?$

Case Study: Portfolio Returns

Portfolio Returns

- Consider the problem of designing portfolios
- The investment manager is considering stocks of Exxon, General Motors, IBM, P&G, Pfizer and FedEx
- He wants to invest $100,000 in these stocks and would like to find the portfolio that is likely to give good returns and not pose too much risk.
- How should he do this?

Collect Data

Date	Exxon Mobil	General Motors	IBM	P&G	Pfizer	FedEx
1/31/1995	15.63	30.56	18.03	16.15	6.81	15.19
2/28/1995	15.97	33.50	18.81	16.49	6.92	16.28
3/31/1995	16.66	34.58	20.53	16.43	7.15	16.91
4/28/1995	17.38	35.47	23.66	17.33	7.22	17.00
⋮	⋮	⋮	⋮	⋮	⋮	⋮
8/31/2004	46.10	41.31	88.15	55.97	32.67	81.99
9/30/2004	48.33	42.48	87.07	54.12	30.60	85.69
10/29/2004	49.22	38.55	84.69	51.18	28.95	91.12
11/30/2004	51.25	38.59	85.74	53.48	27.77	95.03
12/31/2004	51.26	40.06	89.75	55.08	26.89	98.49

Convert to monthly returns

Compute Covariance and Correlation

Covariance	Exxon	GM	IBM	P&G	Pfizer	FedEx
Exxon	0.0021	0.0011	-0.0001	0.0005	0.0005	0.0010
GM	0.0011	0.0086	0.0000	0.0008	0.0053	0.0010
IBM	-0.0001	0.0000	0.0097	0.0001	0.0012	0.0013
P&G	0.0005	0.0008	0.0001	0.0046	0.0030	0.0006
Pfizer	0.0005	0.0053	0.0012	0.0030	0.0721	0.0060
FedEx	0.0010	0.0010	0.0013	0.0006	0.0060	0.0080

Correlation	Exxon	GM	IBM	P&G	Pfizer	FedEx
Exxon	1.0000	0.2513	-0.0313	0.1500	0.0379	0.2318
GM	0.2513	1.0000	-0.0001	0.1263	0.2110	0.1197
IBM	-0.0313	-0.0001	1.0000	0.0202	0.0437	0.1435
P&G	0.1500	0.1263	0.0202	1.0000	0.1648	0.0992
Pfizer	0.0379	0.2110	0.0437	0.1648	1.0000	0.2496
FedEx	0.2318	0.1197	0.1435	0.0992	0.2496	1.0000

Behavior of returns in future …

- **Assume returns will be random from bell-shaped distribution**
  - **reasonable as long as mean is significantly bigger than std. deviation (typical for short-term analysis)**
- **Assume that the correlations remain as before**
  - **caveat: history may not be enough to predict future**
- **Then, find the portfolio that maximizes return while not incurring "too much risk"**

Modern Portfolio Theory (Mean-Variance Analysis)

Let $w_1, w_2, w_3, w_4, w_5, w_6$ represent the weights of Exxon, GM, IBM, P&G, Pfizer, and FedEx in the portfolio, respectively.

2. Objective Function:

Maximize the expected portfolio return:

$$\text{Maximize } R_p = w_1R_1 + w_2R_2 + w_3R_3 + w_4R_4 + w_5R_5 + w_6R_6$$

where R_p is the expected return of the portfolio, and R_i is the expected return of stock i for $i = 1, 2, 3, 4, 5, 6$.

3. Risk (Variance) Constraint:

Ensure the portfolio risk is below a specified threshold $\sigma^2_{\max}$:

$$\text{Variance } (\sigma_p^2) = \sum_{i=1}^{6}\sum_{j=1}^{6} w_i w_j \sigma_{ij} \leq \sigma^2_{\max}$$

where σ_{ij} is the covariance between stock i and stock j.

Modern Portfolio Theory (Mean-Variance Analysis)

$$\text{Maximize } R_p = w_1R_1 + w_2R_2 + w_3R_3 + w_4R_4 + w_5R_5 + w_6R_6$$

Subject to:

$$\sigma_p^2 = \sum_{i=1}^{6}\sum_{j=1}^{6} w_i w_j \sigma_{ij} \leq \sigma^2_{\max}$$

$$\sum_{i=1}^{6} w_i = 1$$

$$w_i \geq 0 \quad \text{for } i = 1, 2, 3, 4, 5, 6$$

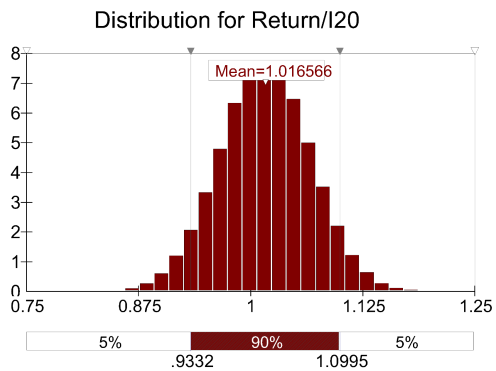
Economist Harry Markowitz introduced MPT in a 1952 essay, for which he was later awarded a Nobel Memorial Prize in Economic Sciences.

https://en.wikipedia.org/wiki/Modern_portfolio_theory

What portfolios might make sense?

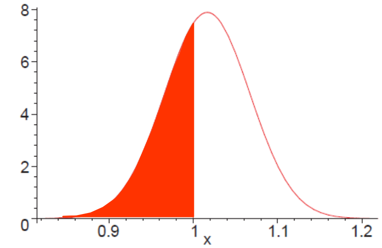
Exxon	GM	IBM	P&G	Pfizer	FedEx	Return	Variance
0.0000	0.0000	0.3490	0.0000	0.0000	0.6510	1.0191	0.0051
0.0000	0.0000	0.3264	0.1681	0.0000	0.5055	1.0180	0.0037
0.1451	0.0000	0.2704	0.2113	0.0000	0.3731	1.0166	0.0026
0.2519	0.0000	0.2371	0.2151	0.0000	0.2958	1.0157	0.0020
0.3160	0.0000	0.2172	0.2174	0.0000	0.2494	1.0151	0.0018

50000 monthly simulations



Return on Investment

- **The portfolio monthly return follows a normal distribution with a mean return of 1.0166 and a standard deviation of 0.05055. What is the probability that the investor will lose money?**



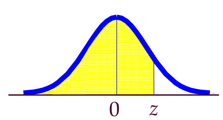
$$P(\text{Monthly rate of return} < 1)$$

$$\text{Standardize 1: } \frac{1-\mu}{\sigma} = \frac{1-1.0166}{0.05055} = -0.3277$$

$$P(R<1) = P(Z<-0.3277) \approx 1-0.6293 = 0.3707$$

Exercise : ROI

P(*Z* <-0.3277) = 1.0 – 0.6293 = 0.3707



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	$\boxed{0.6293}$	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.698[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
0.6	0.7257	0.7291	0.732[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
0.7	0.7580	0.7611	0.764[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
0.8	0.7881	0.7910	0.793[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
0.9	0.8159	0.8186	0.821[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
1.0	0.8413	0.8438	0.846[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319

= Norm.s.dist(-0.3277)

or

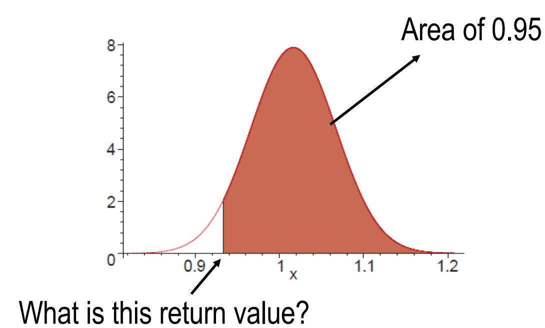
= Norm.dist(1,1.0166,0.5055,1)

Return on Investment

- **In 95% of the cases, the return on investment will exceed what value?**

Find x such that $P(\text{Monthly rate of return} > x) = 0.95$

$$P(R > x) = 0.95$$



Return on Investment

- **Convert the statement in terms of standard normal distribution**

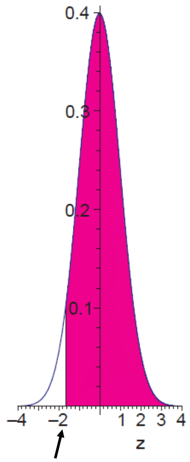
Find the z such that standard normal variable is larger than z with 95% probability

$z_{0.05}$: Look inside normal table for 0.95

$z_{0.05} \approx 1.645$

So, we are looking for -1.645

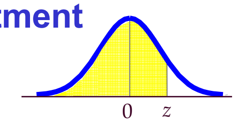
What is the corresponding return?



What is the z-value? $-z_{0.05}$

$$R = \mu + (-z_{0.05})\sigma = 1.0166 - 1.645 \times 0.05055 = 0.9334$$

Exercise : Return on Investment



- **P($Z \leq z$) = 1 – P($Z > z$) = 1 – 0.95 = 0.05**
  $z_{.95}$ = NORM.S.INV(0.05) = −1.645
- **Or from the table: P($Z < z$) = 0.95 → $z_{.05} \approx 1.645$ → $z_{.95} \approx -1.645$**

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	$\boxed{0.9495}$	$\boxed{0.9505}$	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706